

Qatar-2b

R-2123

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_120226 · created 2026-08-11T01:19:43+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455983.95088 +/- 0.00046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.121 +/- 0.025
Transit depth (Rp/Rs)^2	1.46 +/- 0.61 [%]
Orbital Inclination (inc)	89.44 +/- 1.39
Airmass coefficient 1 (a1)	0.6705 +/- 0.0092
Airmass coefficient 2 (a2)	0.3 +/- 0.17
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	#1 - [484, 225]
Optimal Aperture	3.33
Optimal Annulus	9.21
Transit Duration (day)	0.0736 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.121 ± 0.025 vs 0.16327 (deviation 1.31σ)
Duration fit vs published	0.0736 d vs 0.0767 d (deviation 1.19σ)
Mid-time O–C	-0.9 min
Depth SNR	3.7
Residual scatter	1.36 %

Light curve

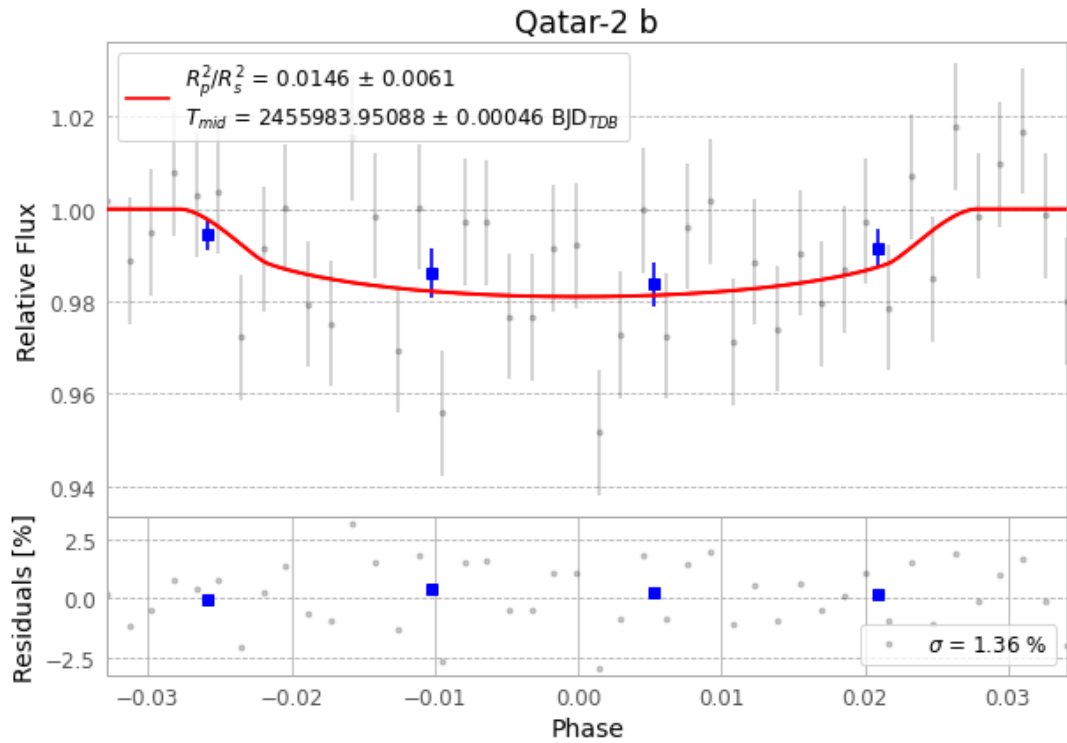


Figure 1 — FinalLightCurve_Qatar-2 b_2012-02-26.png (SHA-256 c746dc9a2a6b6695...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [270, 298], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 785508518c18d52d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

44 FITS frames (29 MB), Qatar-2120226093912.FITS → Qatar-2120226114812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-120226.log (2d8fa36d6913...), FinalLightCurve_Qatar-2 b_2012-02-26.png (c746dc9a2a6b...), FinalLightCurve_Qatar-2 b_2012-02-26.csv (6917c67d955e...)

Compute resources

Wall time 182.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 01:19Z.